

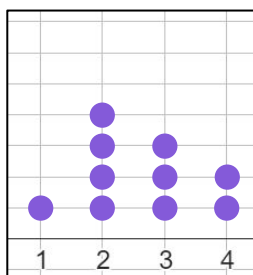


Changing a data point

Task A: How a changed data point influences the mode

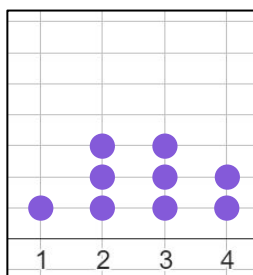
1) Add one point to change the mode for each of the following:

a)



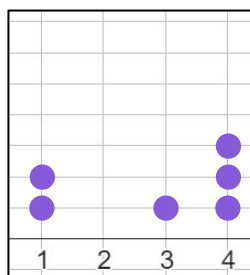
Original mode: 2
New mode: 2, 3

b)



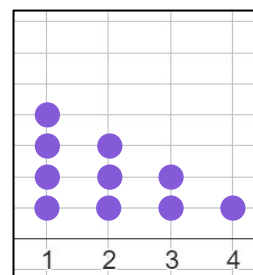
Original mode: 2, 3
New mode: 3

c)



Original mode: 4
New mode: 1, 4

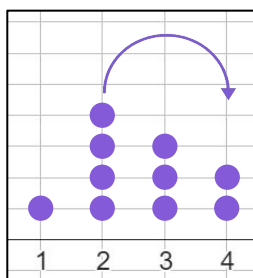
d)



Original mode: 1
New mode: 1, 2

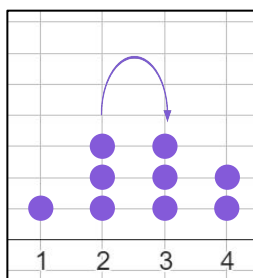
2) Modify one point with an arrow to change the mode for each of the following:

a)



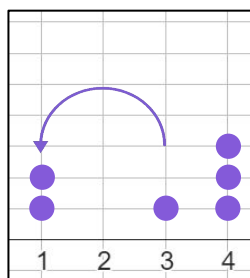
New mode: 2, 3, 4

b)



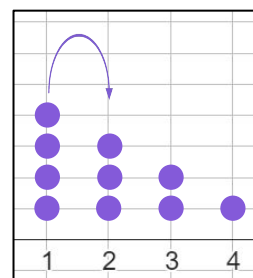
New mode: 3

c)



New mode: 1, 4

d)



New mode: 2



Task B: How a changed data point influences the mean

1) The mean of 5 numbers is 4

a) What is the total value of this data set?

$$5 \times 4 = 20$$

b) If I had to modify one data point from 3 to 8, by how much would the mean increase?

Total increases by 5

$$25 \div 5 = 5$$

The mean increases by 1

c) One extra data point is added to the original data set. The mean stays the same. What is the value of this extra data point?

There are now 6 numbers in the data set.

$$6 \times 4 = 24$$

The value of the new data point is 4

d) One data point is removed from the original data set. The mean decreases to 3. What is the value of the data point that was removed?

There are now 4 numbers in the data set.

$$4 \times 3 = 12$$

The value of the data point that was removed is 8

Task C: How a changed data point influences the median

1) Match the modified datasets with the changed median description. There is one odd one out.

14	14	14	16	16	16
----	----	----	----	----	----

data point of 17 added

14	14	14	16	16	16
----	----	----	----	----	----

data point of 16 removed

14	14	14	14	16	16
----	----	----	----	----	----

data point of 14 modified to 16

median becomes 14

median becomes 15

median becomes 16

median becomes 17



2) Fill in the blanks that fully describe how each of these medians have changed.

89	93	93	97	101	105	109	109
----	----	----	----	-----	-----	-----	-----

data point of 109 removed

The median used to have a value of 99, but has been shifted left by half a data point(s), and now has a value of 97.

44	45	49	58	66	73	80	86
----	----	----	----	----	----	----	----

data point of 73 changed to 48

The median used to have a value of 62, but has been shifted left by 1 data point(s), and now has a value of 53.5.

3) Fill in the blanks that fully describe how each of these medians have changed. You must rewrite the list of numbers with the updated values before completing each sentence.

89	93	93	97	101	105	109	109
----	----	----	----	-----	-----	-----	-----

data point of 101 modified to 95

89	93	93	95	97	105	109	109
----	----	----	----	----	-----	-----	-----

The median used to have a value of 99, but now has a value of 96.

44	45	49	58	66	73	80	86
----	----	----	----	----	----	----	----

data point of 66 removed

44	45	49	58	73	80	86
----	----	----	----	----	----	----

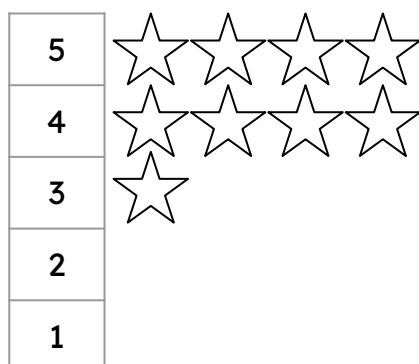
The median used to have a value of 62, but now has a value of 58.

4) Two new products have been reviewed 9 times, each rated on a scale of 1 star to 5 stars.

Each star below represents a review for that product.

What could the 10th ratings of each product be in order for their median rating to increase?

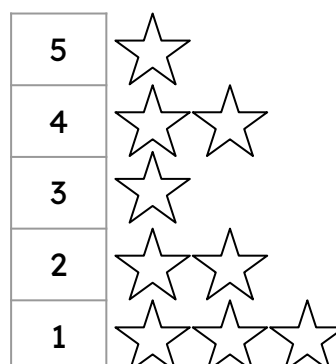
product A



Median rating of product A was 4.

10th rating needs to be 5.

product B



Median rating of product B was 2.

10th rating needs to be 3 or above.

